

Theory Assignment 3

24M0319, 24M0326

Problem Statement:

Find the MAP estimate of μ and σ for $X \sim N(\mu, \sigma)$.

where $\mu \sim N(\mu_1, \sigma_1)$, $\sigma \sim N(\mu_2, \sigma_2)$

Sol:

To find the MAP estimate of μ and σ we need to maximize

$$f(\mu, \sigma | x)$$

Now, $f(\mu, \sigma | x)$

$$\propto f(x | \mu, \sigma) f(\mu, \sigma)$$

$$= f(x | \mu, \sigma) f(\mu) f(\sigma)$$

$\therefore \mu$ and σ are independent.

$$f(x | \mu, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{(x_i - \mu)^2}{2\sigma^2} \right]$$

$$f(\mu) = \frac{1}{\sqrt{2\pi}\sigma_1} \exp \left[-\frac{(\mu - \mu_1)^2}{2\sigma_1^2} \right]$$

$$f(\sigma) = \frac{1}{\sqrt{2\pi}\sigma_2} \exp \left[-\frac{(\sigma - \mu_2)^2}{2\sigma_2^2} \right]$$

Now

$$\ln f(\mu, \sigma | x) \propto \ln f(x | \mu, \sigma) + \ln f(\mu) + \ln f(\sigma)$$

$$= -\frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 - \frac{1}{2\sigma_1^2} (\mu - \mu_1)^2 - \frac{1}{2\sigma_2^2} (\sigma - \mu_2)^2$$

$$\ln f(\mu, \sigma | x) \propto -L$$

$$L = \frac{n}{2} \ln \sigma^2 + \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 + \frac{1}{2\sigma_1^2} (\mu - \mu_1)^2 + \frac{1}{2\sigma_2^2} (\sigma - \mu_2)^2$$

$$\text{So } \max_{\mu, \sigma} \ln f(\mu, \sigma | x) \equiv \min_{\mu, \sigma} L$$

$$\therefore \frac{\partial L}{\partial \mu} \Big|_{\hat{\mu}, \hat{\sigma}} = 0$$

$$\Rightarrow -\frac{1}{\hat{\sigma}^2} \sum_{i=1}^n (x_i - \hat{\mu}) + \frac{1}{\hat{\sigma}_1^2} (\hat{\mu} - \mu_1) = 0$$

$$\Rightarrow \frac{1}{\hat{\sigma}^2} \sum_{i=1}^n x_i - \frac{n\hat{\mu}}{\hat{\sigma}^2} + \frac{1}{\hat{\sigma}_1^2} \hat{\mu} - \frac{\mu_1}{\hat{\sigma}_1^2} = 0$$

$$\Rightarrow \left(\frac{n}{\hat{\sigma}^2} + \frac{1}{\hat{\sigma}_1^2} \right) \hat{\mu} = \frac{1}{\hat{\sigma}^2} \sum_{i=1}^n x_i + \frac{\mu_1}{\hat{\sigma}_1^2}$$

$$\Rightarrow \hat{\mu} = \frac{\frac{1}{\hat{\sigma}^2} \sum_{i=1}^n x_i + \frac{\mu_1}{\hat{\sigma}_1^2}}{\frac{n}{\hat{\sigma}^2} + \frac{1}{\hat{\sigma}_1^2}}$$

$$\hat{\mu} = \frac{\sigma_1^2 \bar{x} + \frac{\hat{\sigma}^2}{n} \mu}{\sigma_1^2 + \frac{\hat{\sigma}^2}{n}} \rightarrow (1)$$

$$\frac{\partial L}{\partial \sigma} \bigg|_{\hat{\mu}, \hat{\sigma}} = 0$$

$$\Rightarrow - \frac{\sum (x_i - \hat{\mu})^2}{\hat{\sigma}^3} + \frac{\hat{\sigma} - \mu_2}{\sigma_2^2} + \frac{1}{\hat{\sigma}} = 0$$

Here there's no closed form solution because of coupling of $\hat{\mu}$ and $\hat{\sigma}$. ~~This so well~~ This problem stems from the problem statement itself.

~~We have seen~~ In the problem statement the assumption is that σ ~~follows~~ has a Gaussian prior. But standard deviation is ^{always} non-negative. Also, ~~if~~ But Gaussian is not a ^{appropriate} ~~proper~~ "conjugate prior" for σ in this problem. Because of this we don't have a closed form solution (but it can be solved numerically). In order to have a closed form solution we will have to consider proper "conjugate prior" for σ .